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Arabic Named Entity Extraction: A Local Grammar-Based Approach

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***Abstract*—The local grammar approach was first used to discuss recursive phrases that are commonly found in specialist literature like biochemistry and then extended to extract time, date and address expressions from letters. It has recently been applied to extract person names from English, Chinese, French, Korean, Portuguese, and Turkish news texts. This paper shows that this approach can also be used to extract person names from Arabic counterparts.**

I. Introduction

THE local grammar approach was first used to discuss recursive phrases that are commonly found in specialist literature like biochemistry and then extended to extract time, date and address expressions from letters. It has recently been applied to extract person names from English, Chinese, French, Korean, Portuguese, and Turkish news texts. This paper shows that this approach can also be used to extract person names from Arabic counterparts[1].

Harris in [2] defines a local grammar as a way of describing syntactic restrictions of certain subsets of sentences, which are closed under some or all of the operations in the language. Frozen expressions may be considered as a subset of sentences that have such syntactic restrictions. One can in fact observe restricted distributions over a number of words. Consider for example:

*Director of (company + thesis + conscience + *chocolate)*
(financial + stock + E) market
*The 20 March (next + 2006 + *bombastic)*

Recall the definition of frozen expressions given in [3]: polylexical units having a frozen characteristic defined through two types of constraints: syntactic and semantic. We can make in this respect some comments regarding our domain – news texts:

- Certain expressions are strictly frozen (e.g. stock market). We can in fact classify these as "compound words" without any loss of generality. On the other hand, others are only partially frozen: they do not accept any complement (one clearly sees semantic constraints), but offer a certain freedom: one could thus talk *about the director of small company, the director of doctoral thesis* and so forth. These are called as semi-frozen expressions.
- For certain expressions such as times, dates and other types of proper names, it appears impossible to individually identify the set of all possible constructions and much more effective a representation in the form of automata. This representation is very natural and can be read easily (if of course the graphs are well arranged). We give an example of the local grammar of some English time expressions in Fig. 1 below. It recognizes expressions like *five in the afternoon*, *5 p.m.*, and *half past one*.

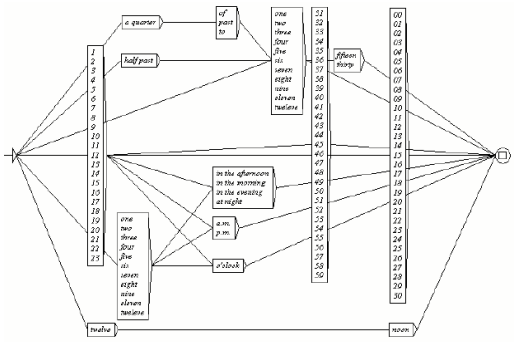

Fig. 1 Example of local grammar of time expressions in English

Local grammars as finite state local automata have been also used to recognize person names in textual documents [4][5][6][7]. This work extends the work concerning the recognition of English person names in [1] to Arabic news texts. In this domain and pretty similar to English, person names were found to occur frequently in constructions having consistent structures in the proximity of Reporting Verbs (RVs) like (**قال**: said, **أضاف**: added, **صرح**: declared, etc.), which are sufficiently frozen to be described in the form of local grammars: they contain slots, which can only be filled in with specific types of linguistic units. An example of these local grammars is shown in Fig. 2 below.

This graph is able to recognize constructions like these:

.... اوبك [ان_ORG [رئيس_ TITLE [أحمد الفهد الصباح]_PN قال
said Ahmad al-Fahd al-Sabah OPEC's president **that**

...." [دلتا لتداول الاوراق المالية]_ORG [من]_AT [محمد رضوان]_ PN **قال**
said Mohamed Radwan of Delta Securities **" ..."**

.... اوبك] **انه**_ORG [ل]_AT [القائم باعمال الامين العام]_TITLE [عدنان شهاب الدين]_ PN **قال**
said Adnan Shihab-Eldin the acting secretary general of OPEC **that**

The grayed boxes labeled PN, ORG, TITLE, AT are the names of sub-graphs that recognize candidates of person names, organization names and job titles respectively. The grayed box AT is to recognize locative prepositions such as (**ل**: of), (**ب**, **في**: in), (**لدى**: with) and so on. Local

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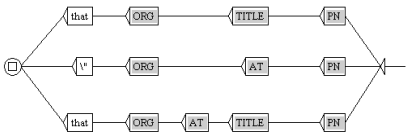

Fig. 2 Example of the local grammars of person names in news articles

grammar graphs containing sub-graphs exhibit similarity to recursive transition networks. Details about their mathematical formalism and features are discussed in details in [1].

The automatic analysis of natural language texts has in recent years covered a range of theoretical and practical enterprises. One key hybrid enterprise is that of Information Extraction (IE). As shown earlier, the IE subtask of identifying proper names in running texts is defined as Named Entity Recognition (NER). In this subtask, finding only names is called name tagging.

Despite the availability of large lexical systems, the analysis of naturally occurring texts is still not an easy task. The difficulty of such analysis is mainly blamed on the ever growing number of new words arising from new coinages, slang expressions, and especially person names. The latter are of particular concern as they fall partly inside and partly outside the lexicon and grammar of the language and are frequent in news texts. Despite the fact that the probabilistic approach (the supervised machine learning)and the symbolic approach (the rule based) have been successful in recognizing Arabic person names in news texts [8],[9],[10] these approaches require large tagged corpora, dictionaries or gazetteers, lists of proper names, which could have been avoided if the local grammar approach was used the way they do. Our local grammar approach is based on three main observations. The first is that the subject argument of the class of verbs known as RVs is a high animacy noun (i.e. it must refer to a person). The second is that the position of the subject argument with respect to the predicate verb is grammatical (i.e. straight after it in Arabic) and therefore predictable. And the third observation is that reporting verbs are highly frequent in news texts.

Having known that person names cluster around RVs in news texts, what is left is to device a method to reveal their local grammar. One way to discover this grammar is to adapt the spirit of the corpus-based method detailed in and goes into three analysis:

1. Frequency analysis
2. Collocation Analysis
3. Concordance Analysis

As words formation in Arabic is different from that of English, the method shown in cannot be directly applied to Arabic texts and thus extra work is needed. Morphologically speaking, Arabic is a non-concatenative language. The main problem with generating Arabic verbal morphology is the large number of variants that can be generated from the same root . For example, the root (قيل: qiyl) can be used to generate 345 different verbal forms including the RV that we are interested in such as the verb (قال: said). This leads to the so called data sparseness problem which adds problems to information extraction tasks in languages other than English. Fortunately, this problem can be solved by applying a morphological analysis prior to the all three analyses as this greatly helps the count of all RVs originated from the same root and later the application of the local grammar.

This paper has two main contributions: the first is to investigate whether or not person names in Arabic news have a local grammar and the second is to reveal the most significant Arabic RVs as there has been no such an analysis for this Language.

II. Method

A. Introduction

This section presents the methodology used for extracting person names from the Arabic financial news by using the local grammar approach.

B. Data Analysis

We have used the untagged arabiCorpus , which is an online set of Arabic corpora that contains portions of textual documents articulated from different sources: one year of Al-Ahram (Egypt - 1999), two years of Al-Hayat (Saudi Arabia) in separate corpora (1996-1997), and a half year each of At-Tajdid (Morocco) and Al-Watan (Kuwait), the Holy Quran, 1001 Nights, several medieval medical and philosophical texts, 8 Egyptian novels, one Egyptian Arabic play, and some EgyptChat data acquired from the Internet, as well as the Penn Treebank news data. Although most of these corpora are only available in combined groups, individual newspapers can be searched separately i.e. according to a specified domain (news, business, sports, etc.). Table I shows the number of words contained in each of these corpora.

Table I.
Number of words in each corpus of ArabiCorpus

Corpus Name	Number of Words
Al-Hayat96	21,564,239
Al-Hayat97	19,473,315
Al-Ahram99	16,475,979
Al-Watan02	6,454,411
Al-Tajdid02	2,919,782
Premodern	912,996
Treebank	598,590
1001Nights	557,908
Modern Literature	403,901
Novels	387,036
Medieval Science	223,249
Egyptian Colloquial	157,099
Quran	84,532

In addition, arabiCorpus offers an advanced searching strategy which we use for the benefit of this work as it allows word/phrase search through the use of part of speech filters which is considered very helpful for the frequency

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analysis required for the acquisition of the local grammar of person names.

As mentioned earlier, the acquired local grammar is eventually implemented into cascades of finite state automata for its readability and speed. We use for this purpose the finite state editor of Unitex text processor of Marnela-Vallée University. The collocation analysis is performed using the Text Analysis tool as explained in .

C. Frequency Analysis

In this analysis we investigate the frequency distribution of the RVs in the biggest amongst news arabiCorpus corpora, Al-Hayst (1996) corpus, only. The list containing these RVs with their English translations follows:

قال : said
أضاف : added
خبر : told
لاحظ : noted
أفاد : reported
اقترح : suggested
علق : commented
ذكر : stated
أعلن : announced
حث : urged
خلص : concluded

The frequency results as found in Al-Hayat (1996) for each of these Arabic RVs are illustrated in Table II.

TABLE II .
FREQUENCY DISTRIBUTION IN AL-HAYAT (1999)

RV	Frequency	Freq/100000 words
قال	104,161	483
أشار	27,388	127
ذكر	21,072	98
أضاف	21,062	97
أعلن	18,206	84
اقترح	3,550	16
علق	2,001	9
خبر	1,500	7
أفاد	1,216	6
خلص	922	4
حث	638	3

It should be noted that each of these frequencies represents the occurrence of all the forms a given RV can have. For example, the frequency results for the verb قال are distributed over 345 different forms such as (34,501) وقال, (15,539) **قال**, (6,476) **وقالت**, (6,476) **يقول**, (5,883) **نقل**, (3,300) **قالت**, (2,944) **ويقول**, etc. Thus, the concordance analysis has to include not only one form, but others that frequently occur too.

D. Collocation Analysis

The collocation analysis is to discover the words (tokens) that frequently collocate with the Arabic RVs. We performed it against a small but mixed portion of Al-Hayat (1996) news corpus because we only needed to know the types and not all the occurrences of the RVs collocations. The results shown in Table III are those for the most frequent RV - **قال**.

TABLE III .
COLLOCATES FOR قال: $\bar{f} = 4.8$ *and* $\sigma = 32$.

Collocate	Left	Right	*freq*	K-score [1]
ان , إن , انه	2,496	1	2,497	78
» [2]	1,539	649	2,188	54
في	1,224	55	1,279	32
من	559	22	581	14
رئيس	537	0	537	13
وزير	267	0	267	8

A closer examination of **قال** collocations shows that this RV mostly collocates with tokens such as the function words (**ان, إن, انه**), the speech mark (»), the prepositions (**في, من**). These collocations, as we show later, are of great importance and must be taken into consideration.

E. Concordance Analysis

This section only shows the concordances of **وقال** - the most frequent form of **قال** RV concatenated with the conjunction و (and). They are generated from Al-Hayat (1996) using arabiCorpus searching tools, which in addition to the frequency information returns, citations (concordances) for the searched word and each of its inflected forms. These citations are ten words length and can be sorted according to the word appearing before/after the concordance word (i.e. **قال**). Below are some examples of these concordances that contain Arabic person and organization names.

وقال عبدالله بوغايف نائب رئيس الوزراء الشيشاني **إن** لدى السلطات معلومات
وقال عبدالله احمد بدوي وزير خارجية ماليزيا **ان** مهاتير وموسى بحثا
وقال عبدالقادر باجمال نائب رئيس الوزراء وزير التخطيط والتنمية «ان اللجنة
وقالت «المؤسسة العربية لضمان الاستثمار» **إنه** على رغم تسلم السلطة الوطنية
وقالت «ام. سي. آي» ومردوخ **انهما** يعتزمان تأسيس شركة مشتركة لبث
وقال «أبو عمار» لـ «الحياة» بعد اللقاء **أنه** اطلع الملك فهد
وقال «أبو مازن» **انه** تم التوصل الى الاتفاق خلال اجتماع استمر
وقال ابراهيم حاجي وهو مسؤول في البنك المركزي التونسي **إن** تجربة
وقال ابراهيم دقش الناطق باسم منظمة الوحدة الافريقية، **ان** الاجتماع الوزاري
وقال ابراهيم غوشة الناطق باسم «حماس» **ان** الفلسطينيين اداروا ظهورهم لاتفاق
وقال ابراهيموفيتش **ان** سارايفو ستظل كما عرفت في تاريخها «ملتزمة التسامح

[1] $K - score = \frac{freq - \bar{f}}{\sigma}$, $\bar{f}$ and σ the average and standard deviation of the frequency of all words/tokens in the corpus.
[2] This the form of the quotation mark used in Al-Hayat newspaper.

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A deep assessment of these concordances shows that person and organization names appear simultaneously in slots that sometimes interspersed by the prepositions (**في**, **من**) to refer to the organization the associated person names work for. We also note that the RV **قال** is used with named organizations. We refer to these expressions, as local grammars. The structures of most frequent local grammars are shown in Table IV.

TABLE IV.
MOST FREQUENT NAMED ENTITIES STRUCTURES (F: FREQUENCY)

CP	LG	F
قال - «...»	**قال** PN TITLE «...»	309
	قال PN: «...»	302
	قال PN TITLE: «...»	107
قال - «ORG»	**قال** TITLE PN AT «ORG»	96
	قال PN AT «ORG»	44
قال - «...»	**قال** TITLE PN TITLE «...»	14
	قال TITLE PN «...»	9
قال - **ان**	**قال** PN ان	593
	قال PN TITLE **ان**	544
	قال TITLE PN TITLE **ان**	81
	قال TITLE PN **ان**	13

The results shown in Table IV for the Arabic saying verb **قال** are quite similar to our findings in for the English one - *said*. These results show that the simultaneous choice of قال and the speech mark intercepted by either a person name, organization name or both is used most widely (939). Less frequent is the choice of **قال** and the function word) **ان** 873). The analysis of these results leads to the construction of a cascade of Finite State Automata (FSA) for **قال** patterns as explained next.

III. BULDING THE FINITE STATE TRANSDUCERS

As mentioned earlier, the local grammar can be implemented into finite state automata/transducers (FSA/FST) using the finite state graph editor of Unitex text processor.

The local grammars in Table IV above show that lists of day names, prepositions, and job titles need to be provided in order to build the sub-graphs illustrated as DATE, AT, and TITLE respectively. The basic challenge lies in the provision of the latter as job titles are often lengthy multiword word expressions and difficult to be accurately acquired automatically. For this reason and the absence of an accurate automatic Arabic multiword extractor, we manually acquired this list of job titles from the concordances used with the concordance analysis. Fig. 3 shows a representative TITLE sub-graph.

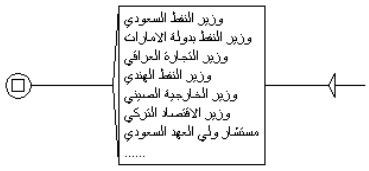

Fig. 3 A portion of TITLE sub-graph

Once the local grammar is implemented into FSTs it can be applied to texts using the *Locate* function provided by Unitex. However, the application of the local grammars is not straight forward. We need first to preprocess the input text so that punctuation anomalies such as the Arabic character *ra* which is sometimes used as a numeric comma or decimal separator, and the Arabic lengthening character used as an *em* dash or numeric hyphen are fixed. The preprocessing stage includes also the splitting of the input text into sentences. A set of finite state transducers like those illustrated in were built for Arabic where the dot and comma is mainly used as sentence delimiter to achieve this goal. Some of the FSA built for Arabic person name local grammars are shown in the Appendix.

IV. AFTERWORD

This paper discusses the use of corpus linguistics methods and techniques, in conjunction with the so-called local grammar formalism, to identify patterns of person names in Arabic news texts. Surprisingly a small vocabulary comprising saying verbs and prepositions together with punctuations and function words like **انه**, **إن**, **ان** and speech marks, are used to acquire local grammars (patterns) of great importance for Arabic proper names extraction.

We tried to show that the application of this method could save us a lot of time and effort needed if all the expectations related to the universal grammars were considered. The small-scale evaluation studies that we had so far are promising. Our long term evaluation involves the application of the discovered local grammar on larger data set to check its accuracy and comprehensiveness.

We believe that although the acquired local grammar patterns do not cover all the names in a given news texts, they help increase dramatically the precision and recall of an Arabic entity extraction system. A study of such a system is currently under way.

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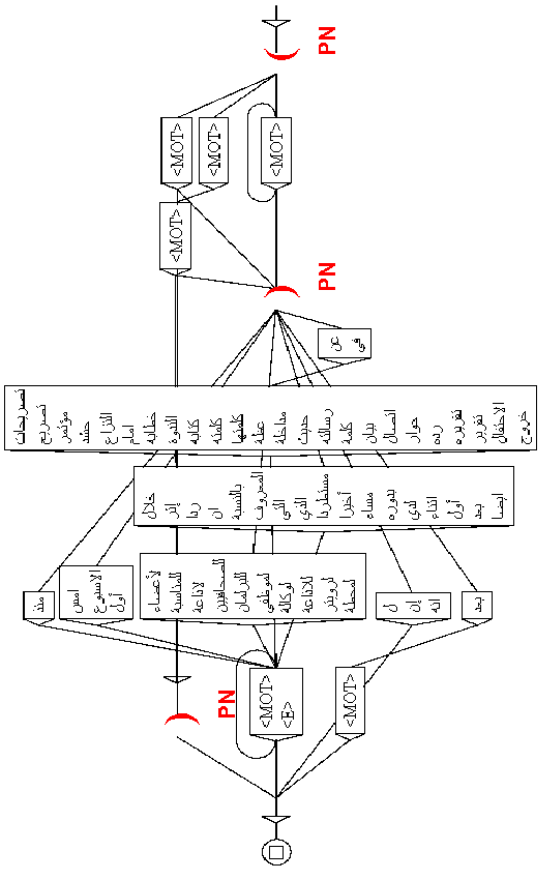
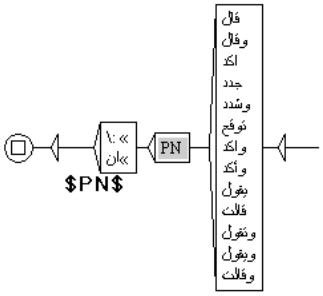
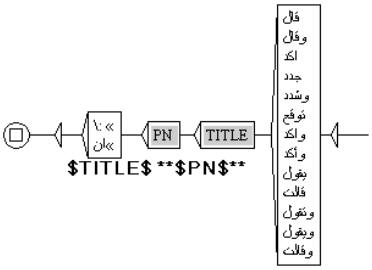
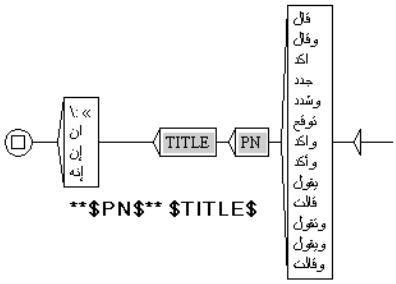
Appendix









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